

Instruction Interpretation & Sequencing (Self-Learning)

Q. Explain the concepts of Instruction Interpretation and Sequencing. How is the control flow of instructions maintained during execution?

****Definition to Remember****

Program
what
execute

1. Instruction Interpretation

The process where the Control Unit determines the meaning of a fetched machine instruction.

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2. Instruction Sequencing

The process of determining the memory address of the *next* instruction to fetch.

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3. Modifying Control Flow

While the default flow is sequential, programs require loops and decisions. Control Flow Instructions alter the sequencing process by changing the PC:

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4. Fetch-Decode-Execute Cycle

The synthesis of both concepts:

1. Fetch

****Must-Write Points (for 10 marks)****

1. **Inst

*****Quick Recall*****

`Interpre

Flow: B